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Scrubaroo Thesis Proposals

Overview of Sui

Sui is a Layer 1 blockchain designed for high throughput, low latency and low transaction costs. Instead of an account-based design, Sui uses an object-centric data model in which every asset is an object with its own identifier, owner and type, which allows independent transactions to be processed in parallel. Smart contracts on Sui are written in Move, a programming language originally designed for the Diem blockchain, whose type system prevents digital assets from being copied or accidentally destroyed. Programmable Transaction Blocks allow many operations to be combined into a single transaction, and zkLogin allows users to access applications with familiar web credentials. Walrus is the decentralized storage network of the Sui ecosystem, designed for storing large binary files. Developers can access resources through the Sui Developer Portal, and the ecosystem is supported by an active developer community and grant programmes.

Links and Repositories

Official site	https://sui.io/
Sui GitHub repository	https://github.com/MystenLabs/sui
Sui documentation	https://docs.sui.io/
Walrus documentation	https://docs.wal.app/

About Scrubaroo

Scrubaroo is a Web3 application developed within the School of Computing at Mediterranean College. It combines deep learning with the Sui blockchain to motivate citizens to clean polluted areas. An administrator creates a cleanup bounty by uploading a photo of a polluted location together with its coordinates. A deep learning model (MobileNetV2 / ResNet50) classifies the photo into three waste-density categories (High, Low, No Waste) and suggests a SUI token reward, which the administrator confirms. A citizen logs in with a Sui wallet, claims the bounty, cleans the area and uploads a photo as proof. Once the administrator approves, a Move smart contract transfers the SUI tokens to the citizen's wallet, and all images and metadata are stored on Walrus. Since no waste-density labels were available, the training labels were generated automatically using K-Means clustering on CNN and CLIP features.

Scrubaroo live application: https://scrubaroo.vercel.app/

Scrubaroo is currently a working prototype. The proposed topics extend it into a fully evaluated system in two directions. The first concerns the artificial-intelligence component: validating the automatically generated labels against human judgement, examining how reliable the confidence of the model is, and studying how administrators collaborate with the AI when assigning rewards. The second concerns the blockchain component: evaluating the transaction cost, latency, security and trust assumptions of the system, analysing possible fraud and abuse scenarios, and protecting the reward process against misuse.

Each topic includes a **theoretical component** (analysis, study, comparison) and a **practical component** (experiments, measurements and implementation on the Scrubaroo system). Communication with the supervisor for questions and ongoing collaboration with the Sui ecosystem are included.

Thesis Topics

1. Human-in-the-Loop Waste-Density Assessment: Validating Pseudo-Labels and Evaluating AI-Assisted Reward Decisions in Scrubaroo

The Scrubaroo classifier was trained on labels generated by K-Means from visual similarity alone. An open research question is whether these clusters correspond to the actual amount of waste in an image, or partly to other visual factors such as the type of scene, since the images come from four different public datasets. In addition, Scrubaroo is a human-AI system in which the model suggests a reward and a person makes the final decision, so it is valuable to understand whether this collaboration improves decisions or leads people to accept incorrect suggestions. The student will build a human-labelled benchmark for Scrubaroo, measure the performance of the model against human judgement, and study how administrators work with the AI.

Theoretical component

The student studies weak supervision and pseudo-labelling, and why models trained on automatically generated labels can appear accurate while learning the wrong concept (shortcut learning and dataset bias). The study covers how annotation reliability is measured through inter-annotator agreement, how the confidence of a neural network can be miscalibrated, and the human-factors literature on trust in automation, overreliance and appropriate reliance on AI advice.

Practical component

The student first writes short annotation guidelines that define High, Low and No Waste with example images. The student and a second annotator, normally the student of the second topic, then label the same set of about 500 images independently of each other, balanced across the four source datasets, and their agreement is measured with Cohen's kappa. About 300 of these images are locked as a test set that is never used for training. On this benchmark the student measures how well the K-Means labels agree with the human labels, checks whether the clusters follow the source dataset rather than the amount of waste, and retrains the classifiers with a proper train, validation and test split over several random seeds, reporting mean and standard deviation. A zero-shot CLIP model is added as a simple reference.

Next, the student measures whether the model's confidence is trustworthy (calibration error and temperature scaling). The student also examines how confidence is used when assigning reward tiers, and designs an approach in which the predicted class sets the reward tier and the confidence decides whether a human must review the case, and simulates different thresholds to find the best balance between administrator workload and errors.

Finally, the student runs a user study with about 16 to 20 participants who act as administrators. Each participant assigns rewards to images once without AI help and once with the AI suggestion and its confidence. The study measures decision accuracy against the human benchmark, time per decision, how often participants follow the AI when it is wrong, their trust in the system, and its usability. This also provides a user-experience analysis of the application. Because the study involves human participants, it follows the research ethics approval procedure of the College before it begins, and it starts with a small pilot.

Main outcomes and further work

The main outcomes of this topic are the human-labelled benchmark with its agreement score, the performance of the model on human labels, the calibration and threshold analysis, and the results

of the user study. The dissertation can be extended further with a label-efficiency experiment showing how many human labels are needed to outperform the automatic labels, a deeper analysis of the user study, and a full literature review.

Required skills: Python, basic PyTorch, basic statistics, and interest in human-computer interaction. No prior blockchain knowledge is needed.

Links

Annotation tool	https://labelstud.io/
CLIP	https://github.com/openai/CLIP

2. Trust, Cost and Fraud Resistance in a Blockchain-Based Citizen Reward System: A Security and Performance Evaluation of Scrubaroo on Sui

Scrubaroo uses Sui smart contracts to distribute rewards and Walrus to store images. This topic evaluates the blockchain component in terms of cost, performance, security and trust. Any system that distributes token rewards must consider possible misuse, for example submitting an old or downloaded photo as proof, using multiple wallets to claim many bounties, claiming bounties without completing the cleanup, or collusion between participants. Since bounties are created and approved by an administrator, the trust assumptions of the system also need to be clearly defined. The student will evaluate the performance and security of Scrubaroo on Sui, develop its threat model, and build and test protections against fraudulent proofs.

Theoretical component

The student studies Sui's object model and Move smart contracts, how transaction fees and storage costs work on Sui and Walrus, and what trust assumptions exist when a system mixes on-chain payments with off-chain AI and human decisions. The study covers threat modelling methods, sybil attacks and the defences used in crowdsourcing and incentive systems, the basics of incentive design for citizen participation, and the privacy implications of storing geotagged photos on public decentralized storage.

Practical component

First, the student measures performance and cost on the Sui testnet. For every smart-contract operation (creating a bounty, claiming it, submitting proof, approving or rejecting) the student records the gas cost and the time to finality over many repeated transactions, together with the Walrus storage cost per image and the time and cost of running the classifier. From these measurements the student estimates the cost of running, for example, one thousand bounties, and discusses how the system would scale on mainnet.

Second, the student writes a structured threat model. It identifies the actors (administrator, citizen, attacker), the assumptions the system makes about each, and the realistic attack scenarios: reused or downloaded proof photos, photos taken at a different location, multiple wallets controlled by one person, collusion between an administrator and a citizen, blocking bounties by claiming them without cleaning, and manipulating the AI with crafted images to obtain a higher reward. For each scenario the student documents how the current system responds and proposes a mitigation. The student also reviews the Move contracts for access-control issues and missing safeguards, such as a time limit on claimed bounties.

Third, the student builds automatic checks for proof photos without training a new model. Perceptual hashing detects reused or near-identical photos, image similarity with CLIP checks whether the proof shows the same scene as the original bounty, and, where the photo contains

location and time metadata, these are compared with the bounty coordinates. To test these checks, the student creates a small real dataset by photographing actual cleanups before and after, blurring any faces or licence plates, and adds fake proofs such as photos of other places, repeated photos and edited photos. The checks are integrated into the administrator panel as warnings, so the administrator stays in control while receiving AI help.

Main outcomes and further work

The main outcomes of this topic are the cost and latency results, a concise threat model with trust assumptions, and the detection rates of the fake-proof checks. The dissertation can be extended further with a prototype defence against multiple-wallet abuse (for example exploring the optional verified identity offered by Sui zkLogin, or an on-chain reputation score), a simulation of how different reward levels affect honest and dishonest behaviour, a plan for moving Scrubaroo to mainnet, and a full literature review.

Required skills: basic Move and Sui development (or willingness to learn), TypeScript or Python, and interest in security. Basic image-processing knowledge helps for the fake-proof checks.

Links

Sui zkLogin	https://docs.sui.io/concepts/cryptography/zklogin
Perceptual hashing library	https://github.com/JohannesBuchner/imagehash

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Collaboration

The two topics are complementary. The students are expected to cooperate where their work overlaps; for example, the student of the second topic can act as the second annotator for the benchmark of the first topic, and both work on the same Scrubaroo codebase and administrator panel. Each dissertation is written individually and covers the student's own contribution.